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Adversarial training for diagnosis based on the electrocardiogram

Making deep learning analysis of the electrocardiogram
more robust

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Abstract

Automated **electrocardiogram (ECG)** analysis using deep neural networks has seen promising results in recent years in various fields such as arrhythmia diagnosis and mortality prediction through age proxies. However, these deep learning models are susceptible to small indistinguishable perturbations that change the model prediction. These models also perform significantly worse when faced with domain shifts when applied to new data distributions not previously trained on.

This thesis investigates how training on worst-case artificial samples through adversarial training can help generate models that can readily be adapted through fine-tuning to new datasets. Investigations are done by conducting robustness and transferability experiments using adversarial training on three different tasks in **ECG** analysis: atrial fibrillation diagnosis, sex classification, and age prediction.

These experiments show that adversarial training creates models that generalize better and can be more efficiently adapted to new datasets with higher performance than non-adversarially trained models. This work thus paves the way for models that perform better on new datasets thus requiring less data for fine-tuning. This enables the use of high-performing models in settings where data is more scarce, thus enabling **ECG** analysis tools suitable in developing countries and smaller hospitals.

The experiments also show that adversarial training improves the robustness of the trained models and that this robustness is retained to a smaller degree when transferred to new settings. This yields models that can withstand smaller directed perturbations without significantly changing their prediction. Thus this work also enables more consistent models, which in turn yields models that are more trustworthy.

Keywords

Deep learning, Adversarial training, Electrocardiogram

Sammanfattning

Analys av elektrokardiogram (EKG) med hjälp av djupinlärning har sett stora framsteg senaste åren, bland annat vid diagnostik av hjärtarytmi, förutsägande av ålder utifrån EKG samt dödlighetsanalyser via åldersregressioner. Dessa modeller är dock känsliga till algoritmer som introducerar mycket små omärkbara förändringar i indatan som får modellen att ge ett så inkorrekt svar som möjligt. Dessutom presterar dessa modeller betydligt sämre när de appliceras i nya miljöer som de inte tränats på.

Denna masteruppsats avser därmed undersöka hur träning på artificiell indata genererad genom adversarial attacks, så kallad *motstridig träning* (eng. adversarial training), kan skapa nya modeller som är så pass allmänna att de motstår både adversarial attacks och effektivt kan appliceras i nya miljöer via finjustering av modellparametrarna. Detta görs genom robusthets- och överförbarhetsexperiment som använder sig av motstridig träning i tre tillämpningsområden för djupinlärning inom EKG-analys: diagnos av förmaksflimmer, könsklassificering och åldersregression.

Experimenten visar att motstridig träning skapar modeller som generaliserar bättre och som kan anpassas till nya dataset och där prestera bättre än motsvarande vanliga djupinlärningsmodeller. Detta masterarbete främjar därför modeller som presterar bättre på nya dataset och därmed kräver mindre data för finjustering. Detta bidrar därmed till högpresterande modeller i miljöer där data är begränsad. Detta kan i sin tur vara användbart i analys av EKG i utvecklingsländer eller mindre sjukhus där data är betydligt mer svårtillgänglig.

Experimenten visar att motstridig träning kraftigt reducerar känsligheten gentemot små, avsiktliga förändringar och därmed skapar robustare modeller och att de även är robustare efter finjustering. Detta skapar modeller som därmed kan motstå små förändringar i indatan utan motsvarande stora förändringar i utdatan. Detta arbete visar därmed hur mer konsekventa modeller kan skapas, vilket i sin tur leder till ökad tillit.

Nyckelord

Djupinlärning, Motstridig träning, Elektrokardiogram

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Arvid Eriksson

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List of acronyms and abbreviations

APGD	Auto-PGD
AUPRC	area under the precision-recall curve
AUROC	area under the receiver operating characteristic curve
CNN	convolutional neural network
ECG	electrocardiogram
FGSM	Fast Gradient Signed Method
PGD	projected gradient descent
SDG	Sustainable Development Goal
UN	United Nations

Chapter 1

Introduction

This chapter describes a shorter background for the relevance of this thesis (Section 1.1), the problem statement and research question (Section 1.2), the purpose of the thesis (Section 1.3), its concrete goals (Section 1.4), its general research methodological approach (Section 1.5), and its delimitations (Section 1.6).

1.1 Background

Cardiovascular diseases account for one-third of the deaths worldwide and the **electrocardiogram (ECG)** is a major tool in their diagnoses. Deep neural networks have recently achieved striking success in diverse tasks such as image analysis and natural language processing, and there are great expectations when it comes to how this technology may improve health care and clinical practice. The last few years saw this technology present promising results in the interpretation of the **ECGs** for cardiovascular diagnosis and stratifying mortality risks and there are high expectations that these tools could enable the early and accurate detection of cardiac diseases [1, 2].

Due to the many interacting components and a large number of parameters, deep neural network models have low robustness meaning classification accuracy, and other key metrics, can often heavily decrease when small perturbations on the input are applied [3]. Models for medical applications need to work in scenarios that are potentially different than they were trained on are sought after. This is especially key as there are many different **ECG** datasets available internationally, where tools developed in one context might be relevant in others.

Adversarial training [4] is an effective approach for training deep learning

models that are more robust to perturbations. Mathematically, adversarial training is formulated as a min-max problem, searching for the best solution to the worst-case attacks. It considers the situation where the training samples have been modified by an adversary. Since training samples are adversarially perturbed it is expected that the model will be more robust when faced with new perturbed samples, as well as generally noisy samples.

An important consequence of adversarial training is that the adversarially trained model tends to produce better representations of the data which generally improves transferability [5]. In a medical setting, this is especially important as exact data distribution will differ between different settings, such as between different hospitals. Adversarial training could thus help provide models that better represent data, such as ECG samples, and show larger generalization capability such that a model can easily be applied across different settings. Such a model would then help save resources as new settings and thus domain shifts would not require the training of new models.

1.2 Problem

The problem approached in this thesis is the adversarial vulnerability in state-of-the-art ECG deep learning models. The intention of this thesis is to promote more robust deep learning models through adversarial training and create models that generalize better, especially when transferred to new settings.

1.2.1 Research question

What effect do adversarial attacks and training have on relevant performance metrics of deep learning models used for different types of 12-lead ECG analysis, specifically diagnosing atrial fibrillation, classifying sex, and predicting age?

The research question is further split into the following sub-questions:

- How vulnerable to performance decreases are convolutional neural network (CNN) ECG diagnosis deep learning models from adversarial attacks, specifically gradient-based attacks such as Auto-PGD (APGD)?
- How will adversarial training using APGD of CNN ECG models impact this performance vulnerability?
- How will adversarial training affect the transferability of state-of-the-art ECG models to other ECG datasets that the models have not been

trained on?

1.3 Purpose

The purpose of this project is to advance the field of deep learning analysis of **ECGs**. This will in turn help further sustainable health, give medical doctors tools for diagnosis assistance, and promote trust and robustness of used models. The higher robustness and generalization capabilities of these models is also a purpose in itself as it will help with access to healthcare for all as large models can more easily be adapted to new datasets.

1.4 Goals

The research question is broken down into the following concrete goals:

- Measure the effects of adversarial attacks using **APGD** on state-of-the-art **ECGs** diagnosis models on key metrics such as **area under the receiver operating characteristic curve (AUROC)** and **area under the precision-recall curve (AUPRC)**.
- Measure how adversarial training of state-of-the-art **ECG** deep learning models improves their robustness to adversarial attacks.
- Measure how adversarial training affects the transferability of state-of-the-art **ECG** deep learning models to new datasets.

1.5 Research Methodology

This thesis will conduct a general empirical study by using public datasets for model training and evaluation, as is common in deep learning. The evaluation will be done using test data from both the same datasets and different datasets than the ones trained on. The effects of adversarial attacks and training on the different generated models will thus be evaluated by obtaining metrics, such as **AUROC**, **AUPRC**, and mean absolute error from the predictions of the models on the test data. This will give an accurate estimate of the effects of adversarial attacks on the different models and allow us to estimate the effects of adversarial training. Bootstrapping will also be utilized to acquire an estimate of the confidence intervals of these metrics. Comparisons of the

different models' performance will then be used to confidently answer the research question.

1.6 Delimitations

The scope of this thesis is adversarial training and how it affects performance metrics such as AUROC and AUPRC with and without adversarial attacks. Transferability to new datasets will also be investigated. The experiments will only utilize CNN model architectures. Other related topics such as investigating other types of models suitable for ECG data analysis or the interpretability of the models will not be considered. Implementation of the resulting model in practice, for example in a real-life medical setting, is also not in the scope of the project. The performance under random noise will also not be investigated as this has no clinical relevance due to already used high-performing noise removal techniques [6].

Furthermore, this thesis will only investigate white-box adversarial attacks where full knowledge of the model parameters is known. This is due to the scope of this thesis being the robustness aspect of adversarial attacks rather than the cybersecurity aspect of securing the model from a potential adversary. Adversarial training will also be the sole defense method investigated, other defense methods such as defensive distillation will not be considered as they have been shown to generally perform worse in most applications [7].

Chapter 2

Background

This chapter describes the current state-of-the-art in deep learning for ECG analysis (Section 2.1). It also describes adversarial attacks (Section 2.2), how improving robustness can be done (Section 2.3), and the corresponding defense method adversarial training (Section 2.4). Key related work are also described here (Section 2.5).

2.1 Deep learning

In the past decade machine learning, and especially deep learning, has seen massive growth due to its performance gains in various fields. It has been used in data-driven decision-making in fields such as healthcare, finance, marketing, and education with great success and is beginning to take a central role in society [8].

Deep learning in turn consists of artificial neural networks which are compositions of functions applied to sets of computational nodes given some parameters. These functions are applied set-wise where such a set is called a layer and where the output of a layer is the input for the next layer. See Figure 2.1 for an example artificial neural network where each layer is the result of a function, usually a composition of a non-linear and linear function, applied on the previous layer.

Deep learning, which is generally seen as artificial neural networks with multiple layers, comprises many different model architectures that are useful for different scenarios. These architectures give a framework for the relationships between each layer and what types of functions are used. For example, the transformer architecture [9] is very efficient in natural language processing while the convolutional neural network architecture is efficient in

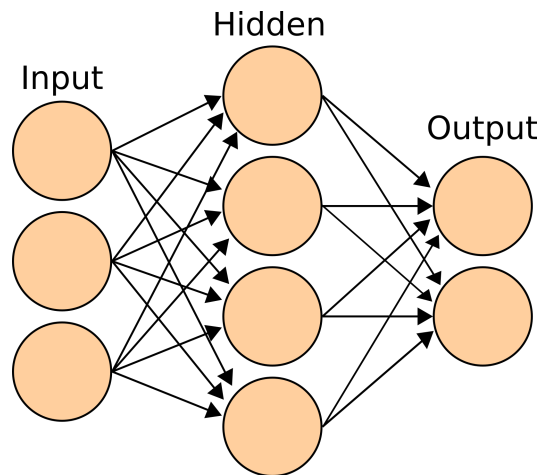


Figure 2.1: An artificial neural network with a single hidden layer, courtesy of Colin M.L. Burnett.

image analysis [10].

2.1.1 Deep learning for the electrocardiogram

Artificial intelligence in general and deep learning especially has seen great success in the automated analysis of the ECG [11]. Deep learning has, for example, been shown to be able to with great accuracy detect left ventricular dysfunction [12], atrial fibrillation [1], and hypertrophic cardiomyopathy [13] as well as predict other features such as clinical age, which in turn can be a predictor for general health [14]. This makes deep learning models promising tools for medical professionals and an important topic for further advancement of ECG analysis.

The reason for deep learning's efficiency in the analysis of the ECG can be attributed to both an abundance of data and suitable model architectures. Since the ECG is a common recording there are several public datasets for such recordings, some of which are available publicly with more than 100 000 unique samples [15, 16]. The amount of data is a key factor for the performance of deep learning models and this relatively high amount of data therefore yields high-performing models. Despite this, some diseases are rare and do not have enough support in actual data to generalize well with and thus require additional methods for regularization.

As model architecture is also vital for the performance of the models another attributing factor is the recent use of convolutional neural networks

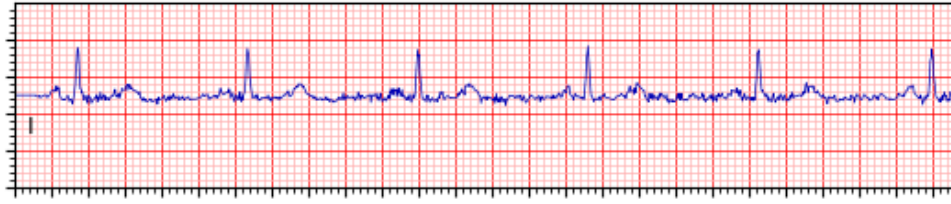


Figure 2.2: An example of an electrocardiogram from the PTB-XL dataset [16].

using residual connections. Convolutional neural networks, which is a type of deep learning architecture, is a state-of-the-art architecture used in deep learning ECG analysis [11]. The suitability of this architecture can be motivated by its translational invariance property which, much like in computer vision, allows the model to find patterns in the data regardless of where in the sequence it is located. This means that the model often will be able to capture important features, such as rapid and irregular beatings signifying atrial fibrillation, independent of at which exact time in the ECG these patterns can be observed.

2.2 Adversarial attacks

While deep learning models have shown striking performance in various number of areas they often also suffer from being sensitive to small perturbations, especially adversarially generated perturbations [17]. These adversarially generated perturbations, or adversarial attacks, are constructed by modifying the input of a model by maximizing the model error under some small perturbation threshold ε which acts as an upper limit of the perturbation. These thresholds are usually so small that they are indistinguishable by a human from real data. Common examples of this are seen in computer vision, where an image of a certain class has its pixels modified by a tiny amount to generate a nearly identical image that the model confidently sees as an entirely different class. For ECGs, such attacks can instead be seen as tiny changes in the signal, often with some type of smoothing to make the perturbations indiscernible from the original [3].

Adversarial attacks can generally be defined as a set of algorithms that seek to, given some knowledge of a model, create an indiscernible change in the data that makes the model misclassify the data. These algorithms can further

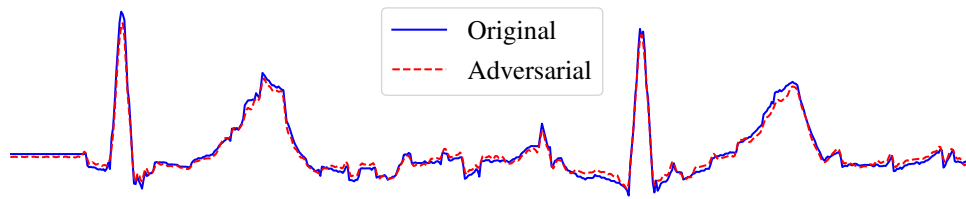


Figure 2.3: Comparison of normal **ECG** and corresponding adversarial example with a perturbation size of $\varepsilon = 0.01$. A normally trained model predicts with Prob > 0.99 that the **ECG** is normal while for the adversarially modified **ECG**, the model predicts atrial fibrillation with Prob > 0.99 .

be divided into black-box and white-box algorithms which assume limited or full knowledge of the model respectively. This thesis focuses on the robustness aspect of adversarial training rather than the cybersecurity aspect and thus white-box algorithms will only be considered.

A common adversarial white-box attack is **Fast Gradient Signed Method (FGSM)** and its corresponding extension **projected gradient descent (PGD)**, both described in detail in Section 3.2. These have seen wide use in the field of adversarial attacks due to their simplicity and efficiency in finding adversarial examples under some maximum perturbation ε [7]. See Figure 2.3 for an adversarial example generated from an **ECG**.

2.2.1 Measuring robustness

As robustness to adversarial attacks often wants to be compared between models suitable metrics for measuring robustness are needed. Large studies have researched different metrics for robustness and their relevance for adversarial attacks [18]. Some such metrics search an entire region around each sample to find an accurate worst-case scenario, however, such methods are extremely costly. A good compromise between finding a worst-case scenario and being computationally feasible is to use the accuracy for adversarial samples generated by **PGD** for some range ε on test datasets from the same distribution as the one trained on [7]. However, this metric also depends on the parameters of the **PGD** which tend to differ in the literature, making comparisons difficult. Efforts have been made to move to versions with perturbation radius ε as the only parameter however these are only recently being adapted through **APGD** [19].

2.3 Improving robustness

Many attempts at improving robustness, loosely defined as consistency between small changes in model input and corresponding output, have been made in the literature in recent years. Some of these methods include detecting and eliminating small changes in the input [20], weakening the impact of changes in the input through different types of data transformation [21], and optimizing the model architecture itself to be robust [22]. However, most of these have quickly been shown to only be robust under certain scenarios and not as general as first thought [23].

The currently most consistent methods for making models robust are instead focused on strengthening the model itself, often through exposure to new types of data. The most prevalent among these is data augmentation, which has seen a huge impact for regularization and robustness in especially computer vision [24]. These techniques generally revolve around manipulating existing data with realistic alterations in order to create more inputs and thus help with the regularization of the model. An efficient and widely used such technique for robustness is adversarial training, where training is done using samples augmented by adversarial attacks. [25].

2.4 Adversarial training

Adversarial training is generally considered to be the most effective defense against adversarial attacks [25, 26, 27] and consists of including samples perturbed by adversarial attacks during training. Adversarial training thus tries to create a robust model that can classify the worst perturbed samples correctly.

In practice, adversarial training consists of adversarially perturbing the input samples during training. An example is adding **FGSM** to the loss function during training, although this has been shown to sometimes lead to a failure mode called *catastrophic overfitting* [28]. While training using **PGD** is slower by a factor equal to the number of steps used, it is still seen as state-of-the-art in adversarial training due to its effect on robustness [7]. However, a problem with **PGD** is the need for hyperparameter tuning since **PGD** takes arguments such as the number of steps, step size, and projection size. There have been attempts to solve this such as using a parameter-free adaptive tuning of the hyperparameters through **APGD** as introduced in [19].

2.5 Related work

Ribeiro et al. [1] shows how residual convolutional neural networks can be utilized for multi-label classification of 12-lead ECGs in order to diagnose several abnormalities such as atrial fibrillation and first-degree atrioventricular block. The work also produces several repositories containing code which is reused in this thesis, mostly for creating similar models.

Han et al. [3] provide an overview of how adversarial attacks can be applied to deep learning in ECG analysis. The work shows that deep learning models for ECG analysis are in general susceptible to adversarial attacks but that most adversarial attacks will create unnatural sequences that can be identified as artificial by a medical professional. They therefore also apply Gaussian smoothing operations on the adversarial examples which retain the examples' adversarial qualities while making the examples indistinguishable from their corresponding true samples. They therefore provide a clear framework for generating adversarial samples for ECG data.

The concept of adversarial training for ECG analysis has been studied before in [29, 30, 31] under various scopes. Hossain et al. [29] investigate using a Conditional Generative Adversarial Network for adversarial training, while Shao et al. [30] attempt combining defensive distillation with adversarial training with promising results. Ma and Liang [31] analyze how adversarial training and some other training regimes provide robustness to adversarial and random noise and implement this using a CNN capable of variable input length. These papers, however, limit their experiments to training and testing on the same dataset and make no transferability experiments. They also base their experiments of robustness on PGD with different hyperparameters. Further advancement in the field could thus be using the parameter-free APGD as well as analyzing the transferability implications of adversarial training. Other techniques for robustness for ECG models have also been attempted, mainly general data augmentation as described by [32]. These augmentations show a slight increase in the performance of the final model.

Chapter 3

Methods

This chapter describes the methods and data used in this thesis. Section 3.1 describes the datasets used in the experiments and how it is preprocessed. Section 3.2 describes how adversarial attacks are implemented and how they are integrated using adversarial training. Section 3.3 describes the deep learning architecture used in this thesis.

3.1 Data

This thesis uses two ECG datasets in its experiments: the CODE dataset [15] and PTB-XL dataset [16].

3.1.1 CODE

The Clinical Outcomes in Digital Electrocardiography (CODE) dataset was developed with the database of digital ECG exams of the telehealth network of Minas Gerais and a detailed description of the cohort can be obtained at [15].

The data set was collected between 2010 and 2017 from 811 counties in the state of Minas Gerais, Brazil, and consists of 2 322 513 12-lead ECG records from 1 676 384 different patients. A subset of 15% of this data set, CODE-15%, is available online [33] and is the variant used for this paper, containing 345 779 samples. This variant will be referred to as CODE for the remainder of this thesis. Ages under 18 are omitted for age prediction.

3.1.2 PTB-XL

PTB-XL [16] is a dataset consisting of 21 799 12-lead ECG records collected between 1989 and 1996 from 18 869 different patients using devices from Schiller AG. It covers a larger range of diagnostic statements than the CODE dataset, with additional types of arrhythmias and diagnostics annotated. Some samples lack notation for age, however, and are omitted in experiments predicting the patient’s age. Ages under 18 are also omitted for age prediction.

Table 3.1: Summary of the datasets. The entries indicate the presence of atrial fibrillation (AF) and distributions of sex and age.

	CODE ($n = 345\,779$)	PTB-XL ($n = 21\,837$)
AF	7 033 (2.0%)	1 514 (6.9%)
Male	139 203 (40.3%)	11 354 (52.1%)
Female	206 576 (59.7%)	10 445 (47.9%)
Ages 18-25	33 096 (9.6%)	838 (3.8%)
Ages 26-40	67 129 (19.4%)	2 105 (9.7%)
Ages 41-60	100 783 (29.2%)	7 193 (33.0%)
Ages 61-80	112 636 (32.6%)	9 137 (41.9%)
Ages ≥ 81	28 155 (8.1%)	2 100 (9.6%)

3.1.3 Preprocessing

The **ECG** signals have been resampled such that all **ECGs** have the same sampling frequency of 400 Hz. Each input **ECG** has 4 096 time samples for each of the 12 standard **ECG** leads. Original signals of a shorter time span have been extended through zero-padding. The samples are also preprocessed through a baseline high-pass filter and a power line notch filter. The class label comprises binary scalar variables corresponding to positive or negative diagnoses for the classification tasks and the predicted age for the regression task.

3.2 Adversarial attacks

In adversarial attacks, an adversary changes the input to the model to maximize the prediction error. It is formulated as a maximization problem, where the

perturbation δ that will be added is computed through the problem

$$\arg \max_{\delta \in S} \mathcal{L}(\theta, x + \delta, y)$$

where $S = \{\delta : \|\delta\|_p \leq \varepsilon\}$, ε is the perturbation radius, and $\mathcal{L}(\theta, x, y)$ is the model loss using model parameters θ for an input sample x and its corresponding true label y .

This section describes **FGSM** and its extension and state-of-the-art adversarial attack **PGD**.

3.2.1 Fast Gradient Signed Method

The **FGSM** is an adversarial attack that seeks to maximize the loss by taking a small step, ε , along the gradient of the loss function $\mathcal{L}$:

$$x_{\text{adv}} = x + \varepsilon \operatorname{sgn}(\nabla_x \mathcal{L}(\theta, x, y)) \quad (3.1)$$

where x is the input, y is the true label, and θ is the model parameters.

3.2.2 Projected gradient descent

PGD is a generalization of **FGSM** where an arbitrary number of steps can be taken instead of **FGSM**'s single step. This is done by iteratively taking steps in the gradient of the loss originating from the previous step:

$$x^{t+1} = \Pi_{x+S}(x^t + \alpha \operatorname{sgn}(\nabla_x \mathcal{L}(\theta, x^t, y))) \quad (3.2)$$

where α is the step size and Π is the projection operator.

An important note here is that since a new loss is calculated each iteration, the final adversarial sample will be costly to generate, having the number of gradient calculations scaling linearly with the number of steps. Doing this for many samples will therefore have a high computational overhead. This is extremely expensive in computer vision, but the cost is manageable with **ECGs** due to the lower dimensionality.

In this thesis, **APGD** is used as an improved version of **PGD** as described in [19] which makes the step size adaptively scheduled during iterations in an explore-exploit-like fashion. **APGD** therefore removes the need to tune the parameter α and therefore only relies on the number of iterations and the boundary S , which in turn only relies on ε .

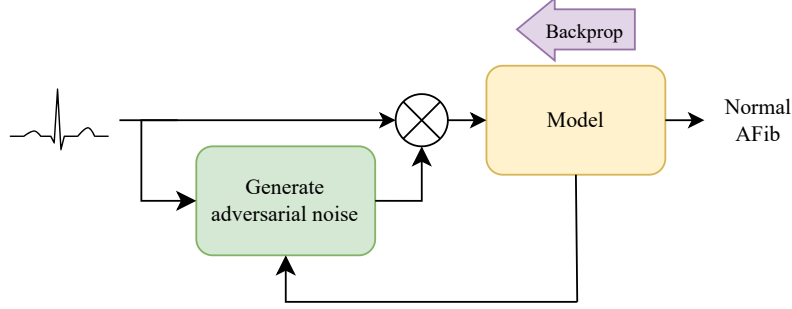


Figure 3.1: Simplified scheme of adversarial training for predicting atrial fibrillation.

3.2.3 Adversarial training

Adversarial training can be formulated as a min-max problem where the training procedure seeks to find the best solution for the worst-case attacks:

$$\min_{\theta} \mathbb{E}_{(x,y) \sim D} \left[\max_{\delta \in S} \mathcal{L}(\theta, x + \delta, y) \right] \quad (3.3)$$

where $(x, y) \sim D$ is the training data sampled from the dataset D and S is the set of allowed perturbations. S is usually dependent on some ε such that $S = \{\delta : \|\delta\|_p \leq \varepsilon\}$ where $p = 2$, representing the euclidean norm, or $p = \infty$, representing the max-norm, are both common choices.

As an estimation of the min-max problem in Equation 3.3, adversarial training is performed by integrating samples generated by adversarial attacks during training. As suggested in [31] the perturbations should ideally be introduced later in the training to avoid early divergence of the model. The perturbation threshold ε of the perturbations can also be scheduled to gradually increase over time to further avoid rapid divergence when the perturbations are introduced. These aspects are integrated during the experiments of this thesis as described in Section 4.1.1. A simplified scheme of adversarial training can be seen in Figure 3.1

3.3 Deep learning architectures

As this thesis seeks to improve the robustness and transferability of state-of-the-art residual **CNNs** a detailed description of their main properties will be described here.

3.3.1 Convolutional layers

Convolutional neural networks are a subset of artificial neural networks suitable for **ECG** analysis where some of the functions applied to layers perform convolutions. These convolutions generally take the form of filters that slide over an image or a time series and create an output that is the input multiplied by the kernel as it slides over the input. See Figure 3.2 for a one-dimensional example similar to the case for an **ECG**. There are usually multiple kernels, which yield multi-dimensional output which is then aggregated.

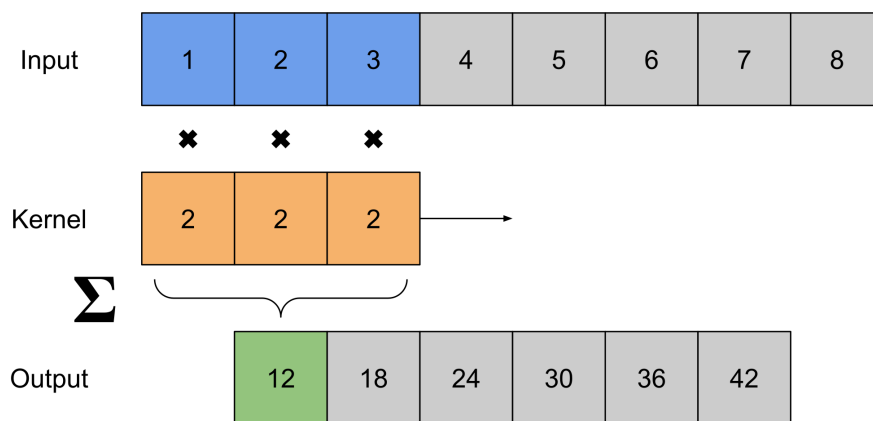


Figure 3.2: Example of a one-dimensional convolutional layer without a bias. The kernel (orange) contains the trainable parameters.

As kernel parameters are the only weights being modified in the layer, and as this kernel is generally small, there is a significant reduction in the number of parameters being optimized compared to a fully connected layer. In Figure 3.2 a corresponding fully connected layer with the same number of input and output nodes would have $8 \times 6 = 48$ weights compared to the 3 used in the convolutional case. This reduction in parameters leads to faster training while introducing a certain bias to the model where spatial features are sought after, thus making the model ideal for computer vision and time series analysis where a pattern at an unknown location in the sequence is sought after, as for **ECG**

analysis. These layers are generally also combined with pooling layers, which help aggregate stacked layers, and residual connections, which speed up the training of the model by introducing skip-connections which skip select layers.

3.3.2 Model architecture

The model architecture used in this thesis is the residual **CNN** proposed in [1] due to its good performance for **ECG** classification and regression. This architecture consists of stacked residual blocks which in turn consist of convolutional layers and residual connections coupled with common deep learning techniques such as batch normalization and dropout. The model's final layers consist of a normal linear (dense) layer followed by a sigmoidal activation in the case of classification which thus outputs an estimate probability for the class being positive. See Figure 3.3 for a detailed schema of the architecture. The architecture used in this thesis is slightly modified, however, by replacing all ReLU activation functions with Gaussian Error Linear Units, GELUs, as these have been shown to increase the robustness of adversarial training [34]. The GELU activation is similar to ReLU but allows for some negative values and a smoother transition for outputs from positive to negative inputs. For a comparison of the two activation functions, see Figure 3.4.

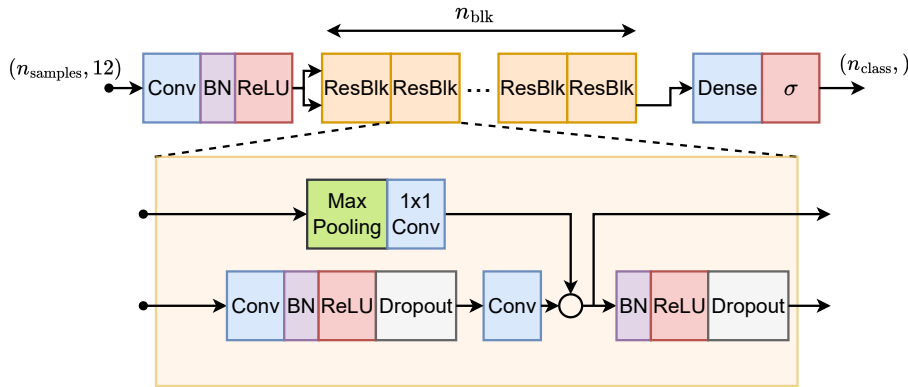


Figure 3.3: The residual convolutional neural network model architecture used in this thesis, replacing the ReLU layers with GELU. (Creative Commons Attribution 4.0 International License, <http://creativecommons.org/licenses/by/4.0/>) [1]

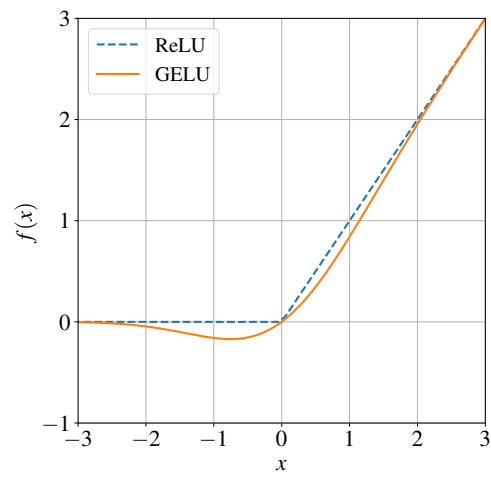


Figure 3.4: Comparison of the ReLU and GELU activation functions.

Chapter 4

Implementation

This chapter details the experiments done in this thesis and the implementation of these. This includes details regarding training such as model training and loss functions (Section 4.1), details of the adversarial robustness experiments (Section 4.2), and details of the transferability experiments (Section 4.3). This chapter also contains the results of the experiments (Section 4.4) and the discussion regarding these (Section 4.5).

4.1 Model training

Models were trained on ECG samples for three different tasks: atrial fibrillation classification, sex classification, and age prediction. All models used the same architecture as shown in Figure 3.3 with the age prediction regression models omitting the final sigmoid layer.

Models were trained using the AdamW optimizer using the recommended optimizer parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1 \times 10^{-8}$, $\lambda = 0.01$ with a learning rate of $\gamma = 1 \times 10^{-3}$ [35]. A batch size of 32 was also used in accordance with [1]. The model parameters were optimized using binary-cross entropy loss for the classification tasks and using mean squared error for the age prediction.

4.1.1 Adversarial training

Some models were also adversarially trained. This adversarial training was performed by training on adversarially modified training samples. Adversarial training was only initialized after an initial warm-up period where the model was trained normally.

Similar to [31] the loss during adversarial training was the average between the standard and adversarial loss where the adversarial loss is defined as $\mathcal{L}(x_{adv}, y)$ where x_{adv} is the corresponding adversarial example generated through 10-iteration APGD [19] with random initialization within the perturbation radius ε under the ℓ_∞ norm. A scaling of the adversarial radius ε using an exponential schedule is also employed using parameters ε_{min} and ε_{max} which represent the minimum and maximum epsilon respectively during scheduling. This lets the model converge before introducing heavily perturbed adversarial examples.

4.2 Robustness experiment

Three PTB-XL models were trained for each of the three tasks (nine models in total) using an 80/10/10 training, validation, and test split. All model cases were trained for 20 epochs. For each task, two out of the three models were adversarially trained. These used a warm-up period of 10 epochs with a common $\varepsilon_{min} = 1 \times 10^{-3}$ and $\varepsilon_{max} = 1 \times 10^{-2}$ and $\varepsilon_{max} = 5 \times 10^{-2}$ respectively and thus a 10 epoch period of adversarial training. The last model for each task was trained without adversarial training as a baseline. See the left-hand side of Tables 4.1 to 4.3 for an overview of these training details.

The robust performance of these was then evaluated by measuring the performance on the test set under 10-APGD adversarial attacks for various ε . For atrial fibrillation the performance is measured using AUPRC, as ROC curves can be optimistic for unbalanced datasets [36], AUROC is used for sex classification, and mean absolute error is used for age prediction.

4.3 Transferability experiment

The models trained in the initial robustness experiments were then fine-tuned on CODE. The fine-tuned models were trained using 10% of the data and tested using 10%, using learning rate $\gamma = 1 \times 10^{-4}$. The low amount of training data was used to simulate a fine-tuning environment where data is more scarce. For atrial fibrillation, a baseline was also normally trained on 90% of the data. The performance of the models for each task was then compared, using a 500-iteration bootstrap and presenting the value and interquartile ranges of the metric.

For the atrial fibrillation classification models the robustness experiment in Section 4.2 is also done for the fine-tuned models and the CODE baseline

model. An attempt at fine-tuning using smaller data sizes was also performed for sex classification in order to compare the performance on even smaller data sizes.

4.4 Results

Figures 4.1 to 4.3 display the results of the robustness experiments. Each figure shows a comparison of three models, a baseline which is normally trained and the two fine-tuned models using $\varepsilon_{max} = 0.05$ and $\varepsilon_{max} = 0.01$ respectively. Note the consistent performance of the adversarial models up until the maximum ε they were trained on and the rapid decrease for larger ε . Also, note the immediate drop in performance for the non-adversarially trained models.

Tables 4.1 to 4.3 show the results of the transferability experiments as the clean performance, i.e. non-perturbed, on the target dataset. The tables show details including training size, length, and time for the original dataset trained on and, if fine-tuning was done, the corresponding data size fine-tuned on and for what size. As all fine-tuning is done for 10 epochs this column has been omitted. Note the very similar performance of the models for sex classification and the larger differences for age prediction and atrial fibrillation classification. The interquartile ranges seem to indicate significant differences in the latter cases. Also, observe that for atrial fibrillation one of the fine-tuned adversarially trained models outperforms a non-adversarially trained model trained for a much higher amount of iterations and more data on the target dataset.

Figure 4.4 shows the robust performance of the fine-tuned models predicting atrial fibrillation. The models here are the same models as the ones shown in Table 4.1, displaying a 95% percentile interval using bootstrap. This shows a clear higher performance of the adversarial models for all ε and is most significant around $\varepsilon = 0.005$.

Figure 4.5 displays the performance comparison for different data sizes fine-tuned when classifying sex. The left-most points are the ones displayed in Table 4.2, while the other points show similarly fine-tuned models but instead using 1% and 0.1% of the dataset for training respectively. No large difference between the models can be observed, but both adversarial models have a higher performance when trained on the smallest dataset (where $n = 346$).

4.5 Discussion

The results in Figures 4.1 to 4.3 clearly show the impact of adversarial training on the robustness of the model. Adversarial training with a higher ε_{max} seems to yield a higher accuracy for higher ε_{max} in exchange for lower clean accuracy and for very low ε_{max} . This same behavior can be seen for all three tasks. It is also interesting to note that the adversarially trained model using $\varepsilon_{max} = 0.01$ for the classification of atrial fibrillation surpasses the clean performance of the base model, this is also in line with [31] which does similar experiments for predicting atrial fibrillation. The results are also in accordance with the robustness accuracy trade-off as described in [37], where a higher ε_{max} yields a more robust model but lower clean accuracy.

Tables 4.1 to 4.3 show that the adversarially trained models seem to be performing better when fine-tuned when compared to their clean performance on PTB-XL (as seen in Figures 4.1 to 4.3). However, the differences in performance between the models seem to vary between tasks. For example, classifying atrial fibrillation (Table 4.1) shows a large performance boost for the $\varepsilon_{max} = 0.01$ adversarially trained model compared to the normally trained and fine-tuned models. Meanwhile, for classifying sex the fine-tuned models show practically identical performance.

A potential reason for the discrepancy between tasks is how the data is balanced for the different tasks. For atrial fibrillation, there are much fewer positive samples than negative ones in both datasets while sex has a similar magnitude in both classes. The latter would thus yield many examples of both negative and positive samples and thus let the fine-tuned models converge faster to a common model, while the atrial fibrillation relies more on the generalization capabilities of the original models. Figure 4.5 strengthens this somewhat by showing that the adversarial models seem to perform slightly better when extremely limited data amounts are available. Though additional experiments are needed to establish the significance of this effect.

Another reason for the differences in performance between the tasks could be the presence of noisy labels for atrial fibrillation. Sex and age are rarely mislabelled in the data while the presence or absence of atrial fibrillation can be misdiagnosed, thus leading to incorrect labels that can severely misdirect the model. Many medical conditions can also be difficult to classify binarily, for example, a patient could have only occasional atrial fibrillation which can make general diagnosis difficult from a single ECG.

As seen in Figure 4.4 the robust performance of the fine-tuned adversarial models is significantly higher than the corresponding base models thus also

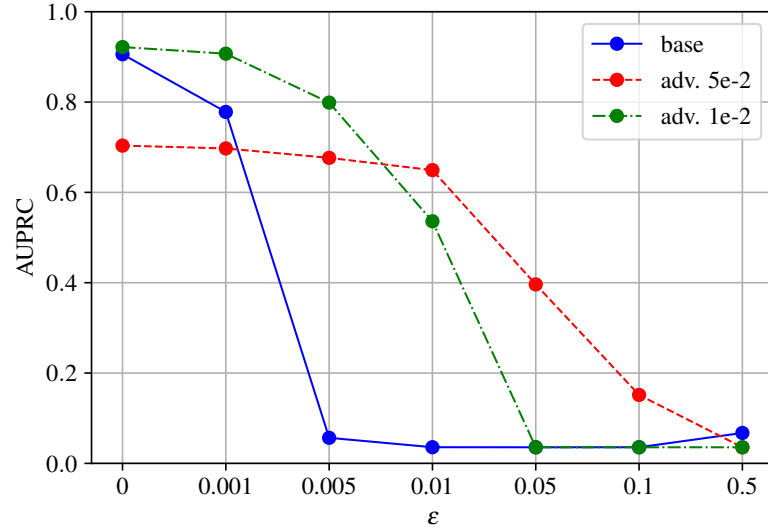


Figure 4.1: Robust performance *classifying atrial fibrillation* on PTB-XL for baseline (base) and two adversarially trained models with $\epsilon_{max} = 0.05$ (adv. 5e-2) and $\epsilon_{max} = 0.01$ (adv. 1e-2) respectively.

signaling that robustness is somewhat transferred when fine-tuning, even when fine-tuning is done through normal training. As expected, since the fine-tuning is done normally and not adversarially, the comparison between Figure 4.4 and Figure 4.1 clearly shows that the robust performance is still significantly worse after fine-tuning. This hints at a relationship between fine-tuning and robustness where fine-tuning’s smaller adjustments of the model parameters transfer some of the original model’s robustness to the new dataset.

Table 4.1: Clean performance of various models *classifying atrial fibrillation* on the CODE dataset measured by **AUPRC**. Under different configurations of training in PTB-XL training and fine-tuning on CODE for 10 epochs.

Training, lr=1e-3					Fine-tuning, lr=1e-4			AUPRC
Dataset	Epochs	Size	Adv. train	t (min)	Dataset	Size	t (min)	
CODE	10	311196	no	99	-	-	-	0.685 (0.675-0.700)
PTB-XL	20	17418	no	15	CODE	34583	10	0.685 (0.673-0.699)
PTB-XL	20	17418	yes, $\epsilon_{max} = 0.01$	40	CODE	34583	10	0.732 (0.721-0.746)
PTB-XL	20	17418	yes, $\epsilon_{max} = 0.05$	40	CODE	34583	10	0.684 (0.674-0.698)

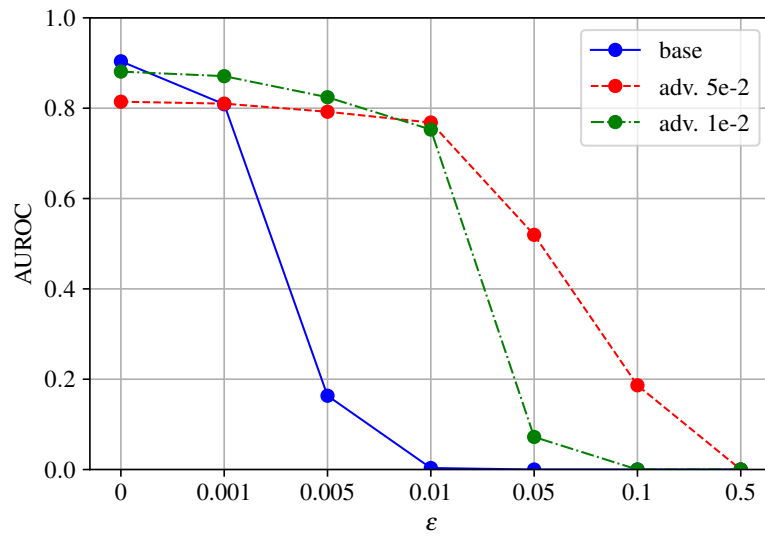


Figure 4.2: Robust performance *classifying sex* on PTB-XL for baseline (base) and two adversarially trained models with $\epsilon_{max} = 0.05$ (adv. 5e-2) and $\epsilon_{max} = 0.01$ (adv. 1e-2) respectively.

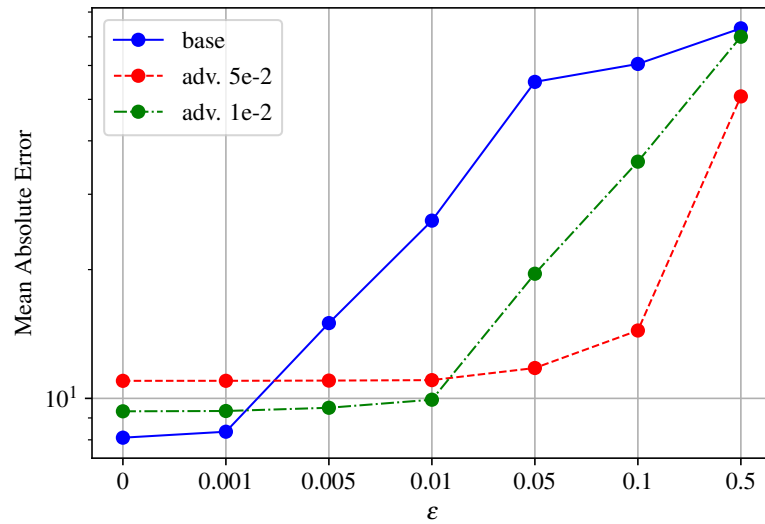


Figure 4.3: Robust performance *predicting age* on PTB-XL for baseline (base) and two adversarially trained models with $\epsilon_{max} = 0.05$ (adv. 5e-2) and $\epsilon_{max} = 0.01$ (adv. 1e-2) respectively.

Table 4.2: Clean performance of various models *classifying sex* on the CODE dataset measured by **AUROC**. Under different configurations of training in PTB-XL training and fine-tuning on CODE for 10 epochs.

Training, lr=1e-3					Fine-tuning, lr=1e-4			AUROC
Dataset	Epochs	Size	Adv. train	t (min)	Dataset	Size	t (min)	
PTB-XL	20	17418	no	15	CODE	34583	10	0.8814 (0.8802-0.8826)
PTB-XL	20	17418	yes, $\varepsilon_{max} = 0.01$	41	CODE	34583	10	0.8809 (0.8796-0.8822)
PTB-XL	20	17418	yes, $\varepsilon_{max} = 0.05$	40	CODE	34583	10	0.8808 (0.8796-0.8819)

Table 4.3: Clean performance of various models *predicting age* on the CODE dataset measured by mean absolute error. Under different configurations of training in PTB-XL training and fine-tuning on CODE for 10 epochs.

Training, lr=1e-3					Fine-tuning, lr=1e-4			MAE (years)
Dataset	Epochs	Size	Adv. train	t (min)	Dataset	Size	t (min)	
PTB-XL	20	17418	no	15	CODE	34583	10	10.33 (10.30-10.36)
PTB-XL	20	17418	yes, $\varepsilon_{max} = 0.01$	39	CODE	34583	10	10.15 (10.12-10.18)
PTB-XL	20	17418	yes, $\varepsilon_{max} = 0.05$	41	CODE	34583	10	9.98 (9.96-10.02)

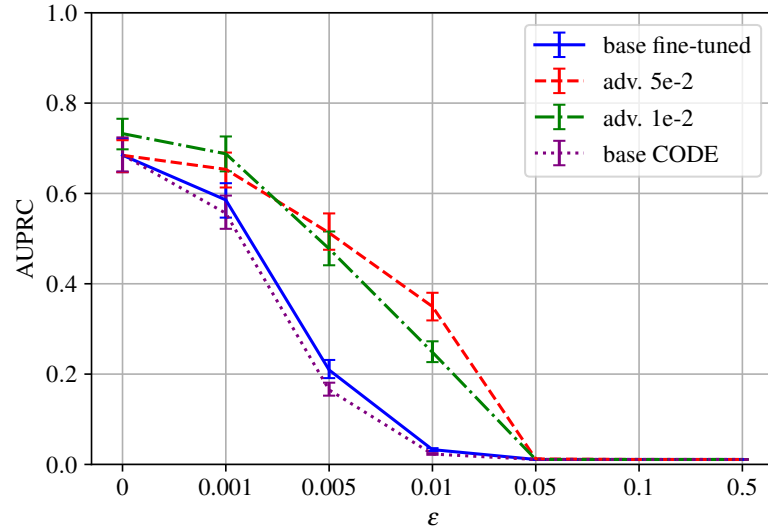


Figure 4.4: Robust performance on CODE *classifying atrial fibrillation* for baseline and the three fine-tuned models in Table 4.1. The error bars show 95% percentile intervals using 500-iteration bootstrap.

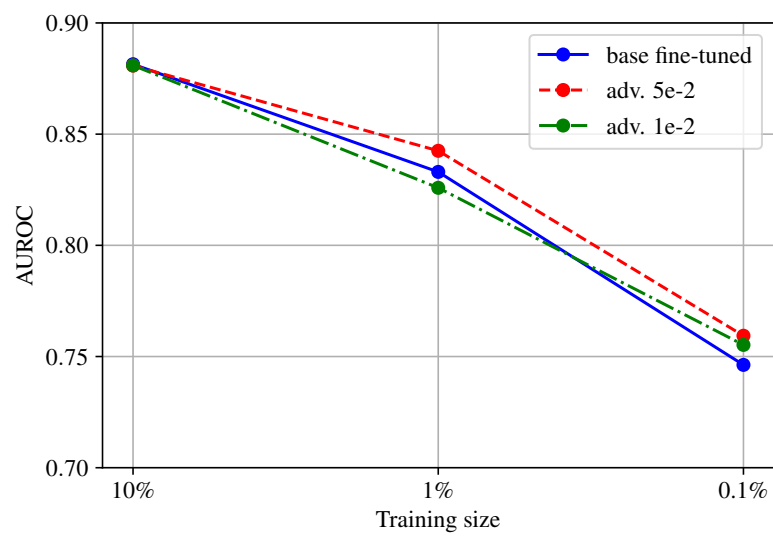


Figure 4.5: Performance on CODE *classifying sex* for fine-tuned baseline and the two adversarial models in Table 4.2 for different sizes of the training set of CODE. All models were fine-tuned for 10 epochs.

Chapter 5

Conclusions and future work

This chapter describes the general conclusions from the experiments done in this thesis (Section 5.1). Furthermore, it discusses limitations and future work (Section 5.2) and sustainability and ethical reflections (Section 5.3).

5.1 Conclusions

This thesis shows that standard deep learning models trained to classify and regress from ECG samples are vulnerable to adversarial attacks and that performance drops rapidly with even very small perturbations. This shows a weakness in existing state-of-the-art models as it indicates a low consistency when practically identical samples can be classified differently and is a vulnerability against potential adversaries that wish to manipulate the models.

When comparing to other results such as [31] the behavior of APGD seems to be much alike that of normal PGD, while also being slightly better at finding worse adversarial examples. The introduction of APGD also allows for easier reproducibility of this work and comparisons.

The results also show that ECG deep learning models can be made robust using adversarial training, which yields an almost consistent performance up until the maximum ε that the adversarial training has used for training. However, the performance drops rapidly for larger ε than that, showing that adversarial training only generalizes well within the perturbation threshold. Another challenge is that the training itself has a high computational cost.

Fine-tuning adversarially trained models also seems to have a higher performance on the new dataset than corresponding non-adversarially trained models, especially when comparing the difference to the performance on the

pre-trained dataset. However, this claim seems to vary between tasks and data distributions, and investigating how the differences between the tasks and data distributions impact the transferability is a potential avenue for future work.

The results also indicate that fine-tuning using adversarially trained models yields a slight increase in performance compared to non-adversarially trained models while also retaining some robustness to adversarial attacks. This suggests that the representations made from adversarially trained models are more suitable for fine-tuning as this yields both increased robustness and performance in the domain transferred to.

5.2 Limitations and future work

The main limitations in this thesis are due to computational costs and a limited number of experiments. Further work should thus ideally promote methods alleviating these challenges and suggested avenues for future research will be made here.

5.2.1 Computational cost

Adversarially training models using several iterations of **PGD** on large numbers of data can be very expensive as the computational time during training increases linearly with the number of iterations. This is a limitation as models suitable for real-life data need to be trained for longer durations to achieve higher performance. This indicates the need for future work to investigate faster forms of adversarial training such as using FastAT [28] or GradAlign [38]. The computational cost of the experiments has also been limiting the amount of models available for the experiments. Statistical analysis using cross-validation would have been valuable, but as these further multiply training times they were outside the realm of feasibility. The number of tasks and datasets investigated also had to be limited due to the computational cost.

5.2.2 Generalizability

The experiments in this thesis have been limited to certain setups using specific models and datasets. As these are limited the generalizability of the claims made here is uncertain, especially those regarding the transferability of adversarially trained models. To further establish these claims larger studies using more datasets should be conducted, to see if the claims generalize

beyond CODE and PTB-XL. Furthermore studying the use of more general models suitable for transfer learning could also be relevant, for example, studying how the multi-label model proposed in [1] is affected by adversarial training in terms of transferability and transfer learning to new tasks. This would also confirm the findings of [5] that adversarial training improves transfer learning.

One could also see adversarial training as a type of data augmentation using the worst possible augments. Therefore it could also be relevant to study that relationship and comparing to data augmentation methods for ECGs such as [32]. Another approach could be to randomly integrate adversarial examples, much like how data augmentation often depends on random additions, to see whether the robustness and transferability of the model improve.

5.2.3 Clinical relevance

Ultimately this work could bring large benefits in clinical settings, especially in developing countries or for smaller hospitals where data sizes are small. For this to be done, however, one needs to further test these models in real-life settings, preferably using adversarially trained versions of general multi-label models such as the one presented in [1]. Furthermore, in a real-life setting, a model should ideally be able to adapt to new incoming data affected by shifts in the distribution caused by changes such as new equipment or routines. This work helps alleviate this issue but does not solve it as performance on new data distributions without fine-tuning is still low.

5.3 Reflections

As this thesis is centered around developing tech to be used in healthcare it mainly contributes to the United Nations (UN) Sustainable Development Goal (SDG) number 3, *Ensure healthy lives and promote well-being for all at all ages*. The SDG targets 3.4, reducing premature mortality, and 3.8, access to universal healthcare, are both especially relevant. This thesis contributes to 3.4 by introducing methods that can assist in predicting early diseases through ECGs. It also contributes to 3.8 by introducing a method for creating general models that can allow high performance on new data. This can be key in developing countries where data might be limited and can thus allow for a cheap way of diagnosing key diseases especially if coupled with cheaper ECG tools such as wearables [39].

This thesis also contributes to cybersecurity as it provides a clear method for making deep learning models resistant to adversarial attacks, which can be generated with malicious intent. This could help prevent scenarios such as insurance fraud where an adversary could generate **ECG** samples using adversarial attacks that would incorrectly diagnose the person. The relevance of this today is, however, extremely limited but could be more relevant in the future as automatic diagnosing becomes more prevalent. This can be seen as a weak link to **SDG** target 16.3, promoting the rule of law.

This thesis also helps alleviate some ethical concerns with machine learning in medical settings. Mainly, as this thesis partly seeks to make deep learning models analyzing **ECGs** robust it helps establish trust with its users as the model will make consistent predictions around some perturbation radius ε . Also, creating more general models by using adversarial training will presumably reduce biases toward dataset-specific features that could be discriminatory in a clinical setting. For example, if a certain ethnicity, gender, or age is overrepresented in the training data of a model then those biases will carry over when applied in a clinical setting. More general models will, however, inherently reduce these biases as they are more general.

The above are also with risks, however. Deep learning models are not perfect and often yield mistakes. Implementing those without supervision in a real-life setting can therefore be dangerous, even if this work increases trust in the models. A risk is therefore too rapid advancement of the field where real lives can be at risk due to too early adaption without preceding clinical studies.

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